



Introduction to Machine Learning (CS 436/536)

Lecture-6: Linear Regression (Cont.)

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General Algorithm

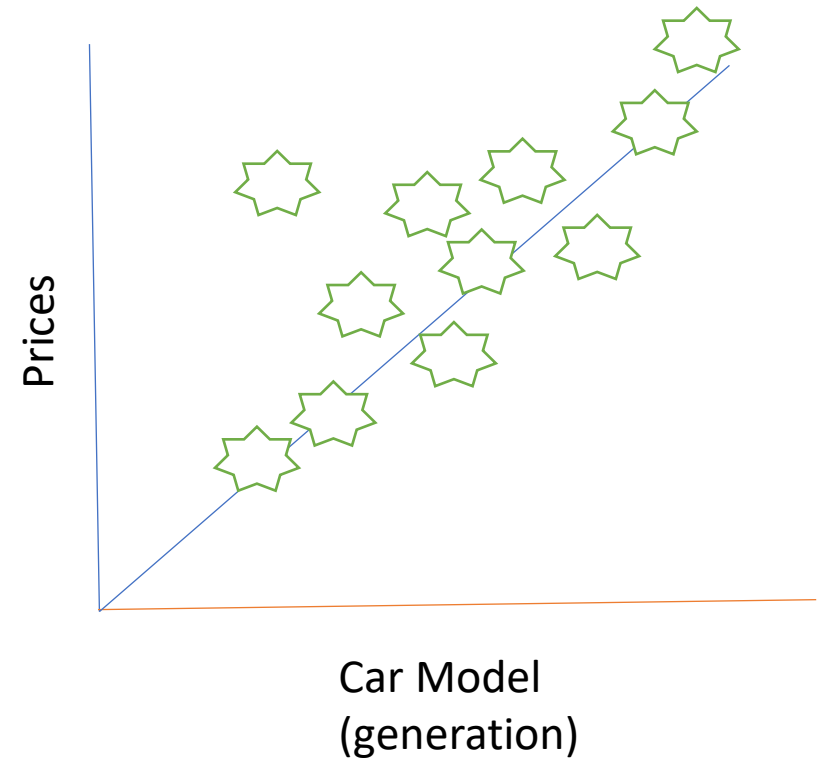
1. Initialize the model parameters θ_0 , θ_1
2. Compute Output $f(x)$ and Objective function J
3. Update $\theta_j = \theta_j - \alpha \frac{\delta J}{\delta \theta_j}$, $j = 0, 1$; α is the learning rate
4. Compute Output $f(x)$ and Objective function J ; until J is minimized.

Regression Problem (Recap):

Car generation (x)	Prices (y)
1	4000
2	10000
3	25000

- Training Data (x_i, y_i is the i^{th} training sample)
- Number of training samples m
- Input Features (x)
- Output values (y)

$$f_{\theta}(x) = \theta_0 + \theta_1 x$$



Applications of Regression:

- House prices (x_1, x_2, x_3)
- Stock prices (x_1, x_2, x_3)
- Exam Scores (x_1, x_2, x_3)
- Rain Forecast (x_1, x_2, x_3)

Applications of Regression:

- House prices (Demand, Size, Income)
- Stock prices (Market, Investors, Economy)
- Exam Scores (Course, Student Prep, Teacher)
- Weather Forecast (Season, Humidity, Temperature)

A New Dataset:

A dataset that includes office rental prices and a number of descriptive features for 10 Dublin city-center offices.

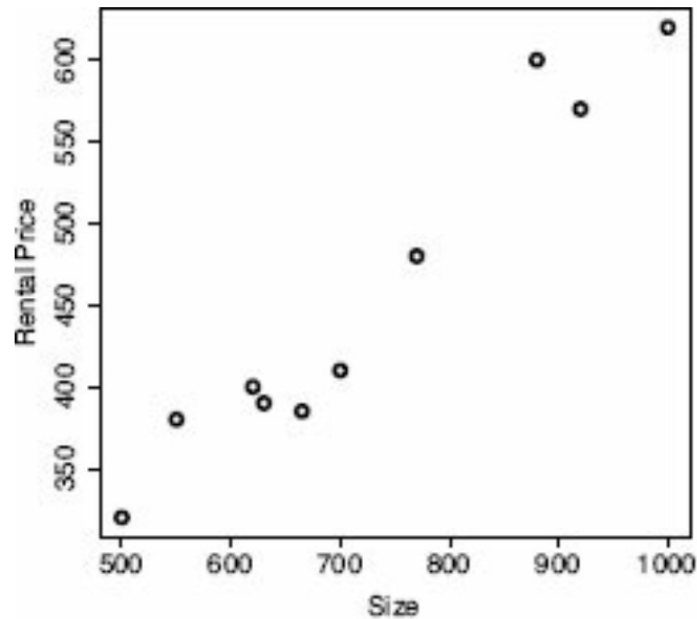
ID	SIZE	FLOOR	BROADBAND RATE	ENERGY RATING	RENTAL PRICE
1	500	4	8	C	320
2	550	7	50	A	380
3	620	9	7	A	400
4	630	5	24	B	390
5	665	8	100	C	385
6	700	4	8	B	410
7	770	10	7	B	480
8	880	12	50	A	600
9	920	14	8	C	570
10	1,000	9	24	B	620

Linear Regression Results:

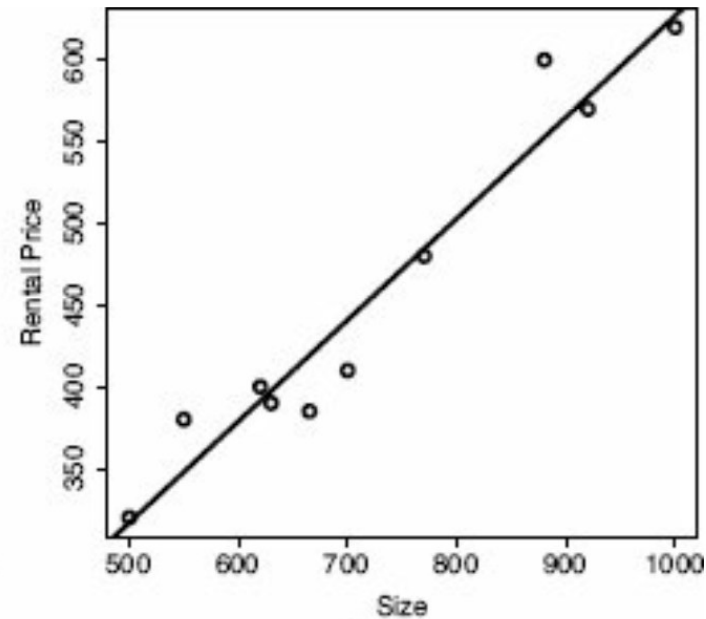
$$\text{RENTAL PRICE} = 6.47 + 0.62 \times \text{SIZE}$$

$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \mathbf{w}[0] + \mathbf{w}[1] \times \mathbf{d}[1]$$

$$y = mx + b$$



(a)



(b)

Notations:

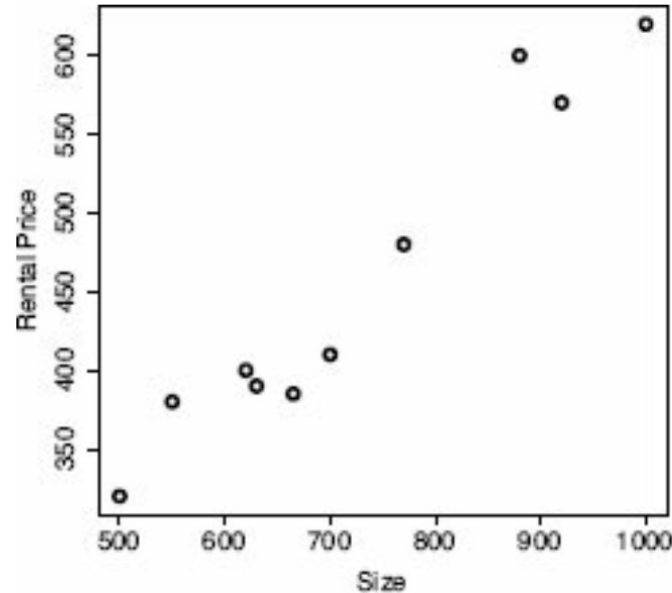
- $W[0] \Rightarrow \theta_0$
- $W[1] \Rightarrow \theta_1$
- $d[0] \Rightarrow 1$ (*in case of LR*)
- $d \Rightarrow x$
- $L2 \Rightarrow J$ (*Error Function*)

Linear Regression Results:

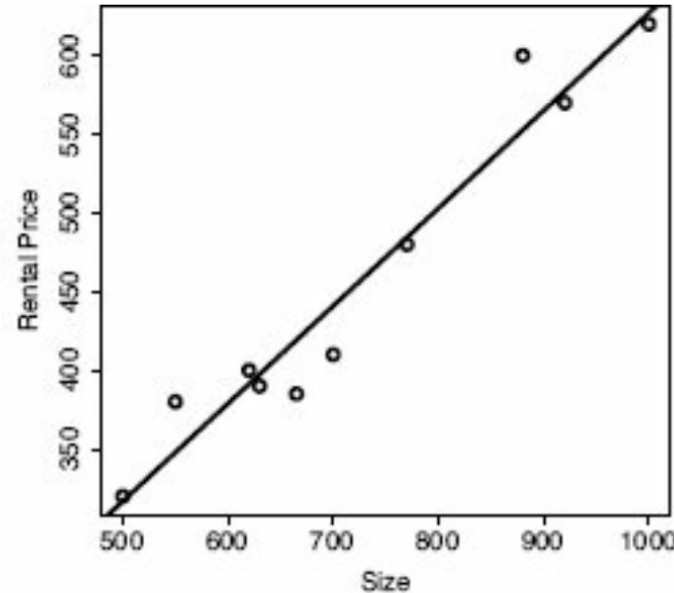
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$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \mathbf{w}[0] + \mathbf{w}[1] \times \mathbf{d}[1]$$

$$y = mx + b$$



(a)



(b)

$$\begin{aligned}\text{RENTAL PRICE} &= 6.47 + 0.62 \times 730 \\ &= 459.07\end{aligned}$$

Dataset:

A dataset that includes office rental prices and a number of descriptive features for 10 Dublin city-center offices.

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Multi-Variable Regression:

$$\text{RENTAL PRICE} = \mathbf{w}[0] + \mathbf{w}[1] \times \text{SIZE} + \mathbf{w}[2] \times \text{FLOOR} \\ + \mathbf{w}[3] \times \text{BROADBAND RATE}$$

$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \mathbf{w}[0] + \mathbf{w}[1] \times \mathbf{d}[1] + \cdots + \mathbf{w}[m] \times \mathbf{d}[m] \\ = \mathbf{w}[0] + \sum_{j=1}^m \mathbf{w}[j] \times \mathbf{d}[j]$$

where $\mathbf{d}$ is a vector of m descriptive features, $\mathbf{d}[1] \dots \mathbf{d}[m]$, and $\mathbf{w}[0] \dots \mathbf{w}[m]$ are $(m + 1)$ weights.

Multi-Variable Regression:

$$\begin{aligned}\text{RENTAL PRICE} = & \mathbf{w}[0] + \mathbf{w}[1] \times \text{SIZE} + \mathbf{w}[2] \times \text{FLOOR} \\ & + \mathbf{w}[3] \times \text{BROADBAND RATE}\end{aligned}$$

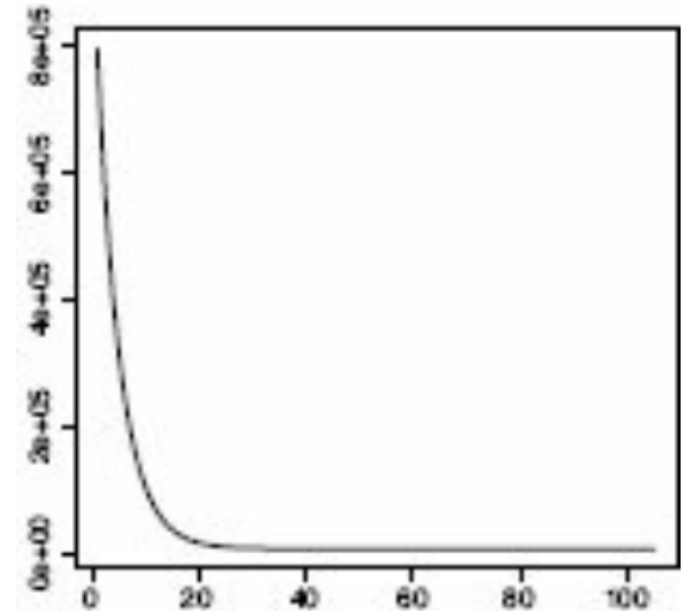
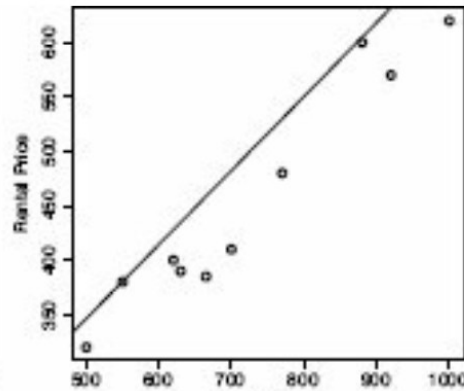
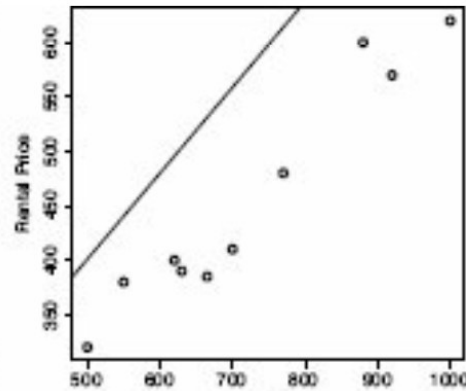
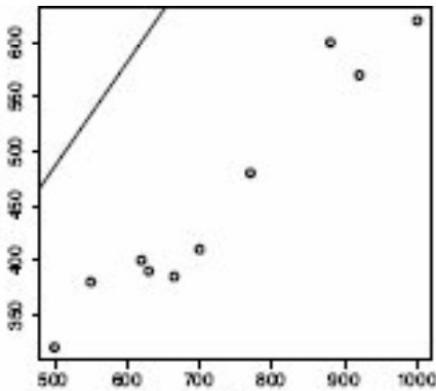
$$\begin{aligned}\mathbb{M}_{\mathbf{w}}(\mathbf{d}) &= \mathbf{w}[0] + \mathbf{w}[1] \times \mathbf{d}[1] + \cdots + \mathbf{w}[m] \times \mathbf{d}[m] \\ &= \mathbf{w}[0] + \sum_{j=1}^m \mathbf{w}[j] \times \mathbf{d}[j]\end{aligned}$$

where $\mathbf{d}$ is a vector of m descriptive features, $\mathbf{d}[1] \dots \mathbf{d}[m]$, and $\mathbf{w}[0] \dots \mathbf{w}[m]$ are $(m + 1)$ weights.

$$\begin{aligned}\mathbb{M}_{\mathbf{w}}(\mathbf{d}) &= \mathbf{w}[0] \times \mathbf{d}[0] + \mathbf{w}[1] \times \mathbf{d}[1] + \cdots + \mathbf{w}[m] \times \mathbf{d}[m] \\ &= \sum_{j=0}^m \mathbf{w}[j] \times \mathbf{d}[j] \\ &= \mathbf{w} \cdot \mathbf{d}\end{aligned}$$

Multi-Variable Regression:

- Error Function:
$$L_2(\mathbb{M}_w, \mathcal{D}) = \frac{1}{2} \sum_{i=1}^n (t_i - \mathbb{M}_w(\mathbf{d}_i))^2$$
$$= \frac{1}{2} \sum_{i=1}^n (t_i - (\mathbf{w} \cdot \mathbf{d}_i))^2$$



Gradient Descent Algorithm:

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times -\mathbf{d}_i[j])$$

Gradient Descent Algorithm:

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times -\mathbf{d}_i[j])$$

$$\begin{aligned} errorDelta(\mathcal{D}, \mathbf{w}[j]) &= -\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) \\ &= \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbf{d}_i[j]) \end{aligned}$$

Gradient Descent Algorithm:

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times -\mathbf{d}_i[j])$$

$$\begin{aligned} \text{errorDelta}(\mathcal{D}, \mathbf{w}[j]) &= -\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) \\ &= \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbf{d}_i[j]) \end{aligned}$$

$$\mathbf{w}[j] \leftarrow \mathbf{w}[j] + \alpha \times \underbrace{\sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbf{d}_i[j])}_{\text{errorDelta}(\mathcal{D}, \mathbf{w}[j])}$$

Gradient Descent Algorithm:

Require: set of training instances $\mathcal{D}$

Require: a learning rate α that controls how quickly the algorithm converges

Require: a function, *errorDelta*, that determines the direction in which to adjust a given weight, $w[j]$, so as to move down the slope of an error surface determined by the dataset, $\mathcal{D}$

Require: a convergence criterion that indicates that the algorithm has completed

1: $w \leftarrow$ random starting point in the weight space

2: repeat

3: for each $w[j]$ in w do

4: $w[j] \leftarrow w[j] + \alpha \times \text{errorDelta}(\mathcal{D}, w[j])$

5: until convergence occurs

Algorithm Intuition:

- If the errors show that, in general, predictions made by the candidate model are too high, then $w[j]$ should be decreased if $d_i[j]$ is positive and increased if $d_i[j]$ is negative.
- If the errors show that, in general, predictions made by the candidate model are too low, then $w[j]$ should be increased if $d_i[j]$ is positive and decreased if $d_i[j]$ is negative.

Working Example:

A dataset that includes office rental prices and a number of descriptive features for 10 Dublin city-center offices.

ID	SIZE	FLOOR	BROADBAND RATE	ENERGY RATING	RENTAL PRICE
1	500	4	8	C	320
2	550	7	50	A	380
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Working Example:

$$\text{RENTAL PRICE} = w[0] + w[1] \times \text{SIZE} + w[2] \times \text{FLOOR} \\ + w[3] \times \text{BROADBAND RATE}$$

For this example, let's assume that the learning rate, α , is 0.00000002 and the initial weights are chosen from a uniform random distribution in the range $[-0.2, 0.2]$ to be $w[0] = -0.146$, $w[1] = 0.185$, $w[2] = -0.044$, and $w[3] = 0.119$.

After one iteration:

Initial Weights

w [0]:	-0.146	w [1]:	0.185	w [2]:	-0.044	w [3]:	0.119
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Iteration 1

ID	RENTAL PRICE	Pred.	Error	Squared Error	<i>errorDelta</i> ($\mathcal{D}, w[j]$)			
					w [0]	w [1]	w [2]	w [3]
1	320	93.26	226.74	51,411.08	226.74	113,370.05	906.96	1,813.92
2	380	107.41	272.59	74,307.70	272.59	149,926.92	1,908.16	13,629.72
3	400	115.15	284.85	81,138.96	284.85	176,606.39	2,563.64	1,993.94
4	390	119.21	270.79	73,327.67	270.79	170,598.22	1,353.95	6,498.98
5	385	134.64	250.36	62,682.22	250.36	166,492.17	2,002.91	25,036.42
6	410	130.31	279.69	78,226.32	279.69	195,782.78	1,118.76	2,237.52
7	480	142.89	337.11	113,639.88	337.11	259,570.96	3,371.05	2,359.74
8	600	168.32	431.68	186,348.45	431.68	379,879.24	5,180.17	21,584.05
9	570	170.63	399.37	159,499.37	399.37	367,423.83	5,591.23	3,194.99
10	620	187.58	432.42	186,989.95	432.42	432,423.35	3,891.81	10,378.16
Sum				1,067,571.59	3,185.61	2,412,073.90	27,888.65	88,727.43
Sum of squared errors (Sum/2)				533,785.80				

New Weights (after Iteration 1)

w [0]:	-0.146	w [1]:	0.233	w [2]:	-0.043	w [3]:	0.121
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Calculation of first line in Previous table:

$$\text{RENTAL PRICE} = w[0] + w[1] \times \text{SIZE} + w[2] \times \text{FLOOR} \\ + w[3] \times \text{BROADBAND RATE}$$

Given, $W[0] = -0.146$, $W[1] = 0.185$, $W[2] = -0.044$, $W[3] = 0.199$

And for $\text{SIZE} = 500$, $\text{FLOOR} = 4$ and $\text{BROADBAND} = 8$

$$Pred = M_w = -0.146 + (500 \times 0.185) + (4 \times -0.044) + (8 \times 0.199) \Rightarrow 93.26$$

$$Error = 320 - 93.26 \Rightarrow 226.74$$

$$Squared Error = 226.74^2 \Rightarrow 51411.08$$

$$Error\ Delta\ for\ W[0] = Error \cdot d[0] = 226.74$$

$$Error\ Delta\ for\ W[1] = Error \cdot d[1] \Rightarrow 226.74 \times 500 \Rightarrow 113370.05$$

$$Error\ Delta\ for\ W[2] = Error \cdot d[2] \Rightarrow 226.74 \times 4 \Rightarrow 906.96$$

$$Error\ Delta\ for\ W[3] = Error \cdot d[3] \Rightarrow 226.74 \times 8 \Rightarrow 1813.92$$

Sample Calculation

- Rental Price = $-0.146 + (500 \times 0.185) + (-0.044 \times 4) + (0.119 \times 8)$
= 93.26

$$W[0] = -0.146 + (0.000000002 \times 3185.61) = -0.146$$

$$W[1] = 0.185 + (0.000000002 \times 2412073.9) = 0.233$$

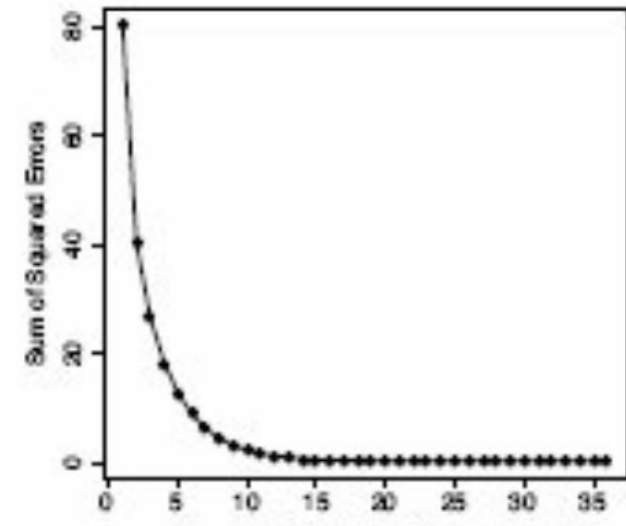
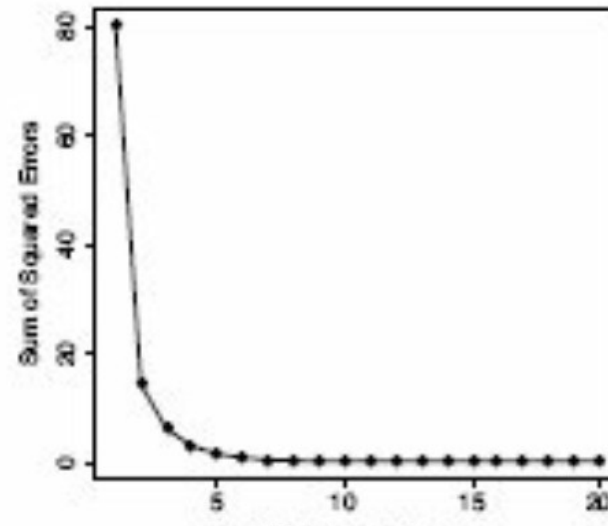
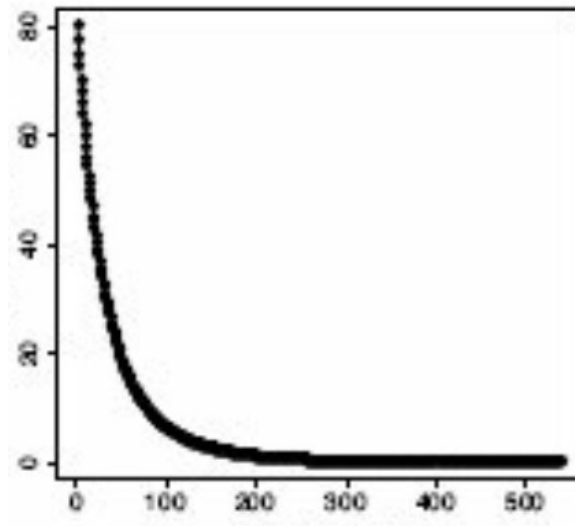
How to Handle Categorical Features:

ID	SIZE	FLOOR	BROADBAND RATE	ENERGY RATING A	ENERGY RATING B	ENERGY RATING C	RENTAL PRICE
1	500	4	8	0	0	1	320
2	550	7	50	1	0	0	380
3	620	9	7	1	0	0	400
4	630	5	24	0	1	0	390
5	665	8	100	0	0	1	385
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Feature Scaling:

- Scale the features
- Normalization
- Mean and Standard deviation of the dataset are μ, σ
- Normalized features: $(x - \mu)/\sigma$
- Normalized features: $(x - \mu)/R$; R is the range of value
- It ensures all the features are centered around mean with smaller values

Effect of Learning Rate:



Dynamic Learning Rate:

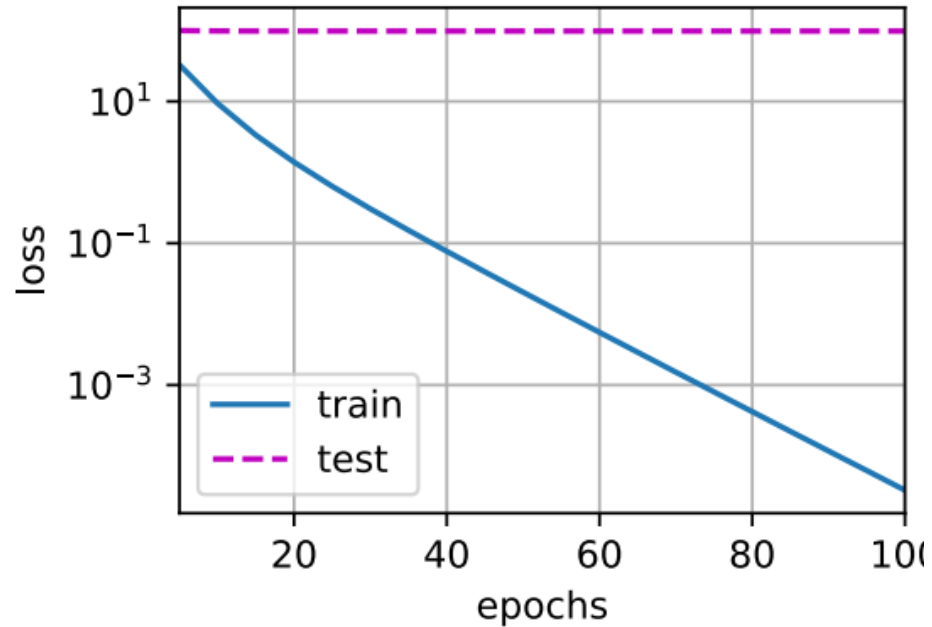
- Smaller learning rate ensures accurate learning with slow convergence.
- It risks falling into a local minima.
- A larger learning rate may not result in convergence.
- Solution is to adopt a dynamic learning rate with iterations.

$$a_{\tau} = a_0 \frac{c}{c + \tau}$$

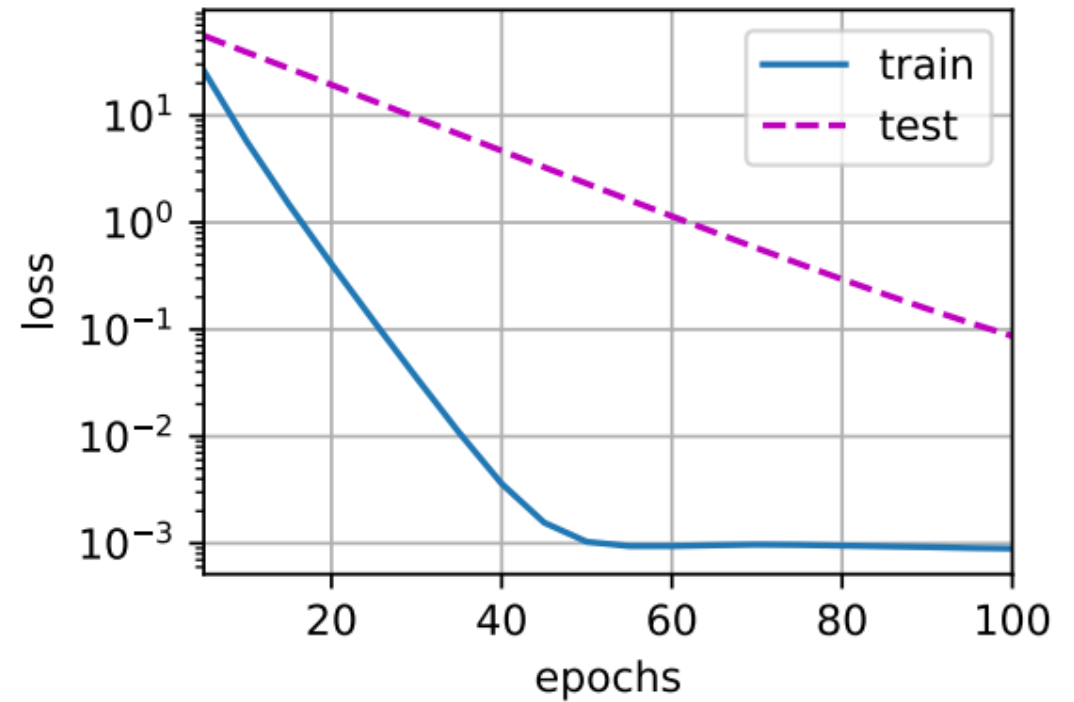
- Here c is constant and τ is the iteration number.

Bias-Variance Trade-Offs

Norm (W) = 12.9



Norm (W) = 0.3



How to Prevent Overfitting

- Reduce Number of Features
- Model Complexity
- Increase Learning Rate
- Keep all features but reduce magnitude (Regularization)
- Dropout
- Increase the Batch Size

Regularization Terms

- Feature selection
 - Lasso Penalty
 - L1 Norm
- Weight Decay
 - Ridge Penalty
 - L2 Norm

One reason to work with the L2 norm is that it places an outsize penalty on large components of the weight vector. This biases our learning algorithm towards models that distribute weight evenly across a larger number of features.

Regularization:

- Lasso Regression

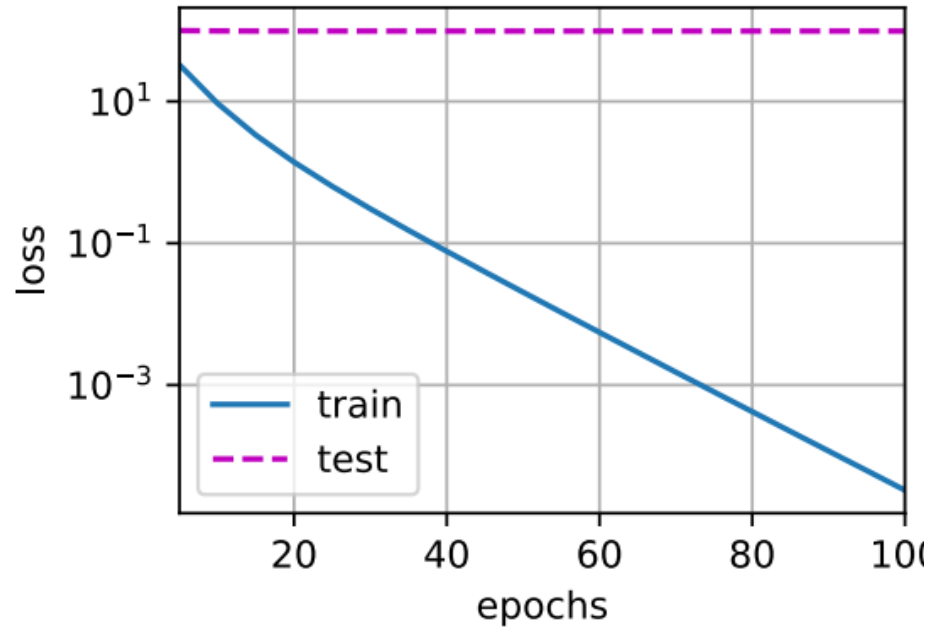
$$\text{Regularized Loss} = \text{Loss} + \lambda * \text{L1 Norm Penalty}$$

- Ridge Regression

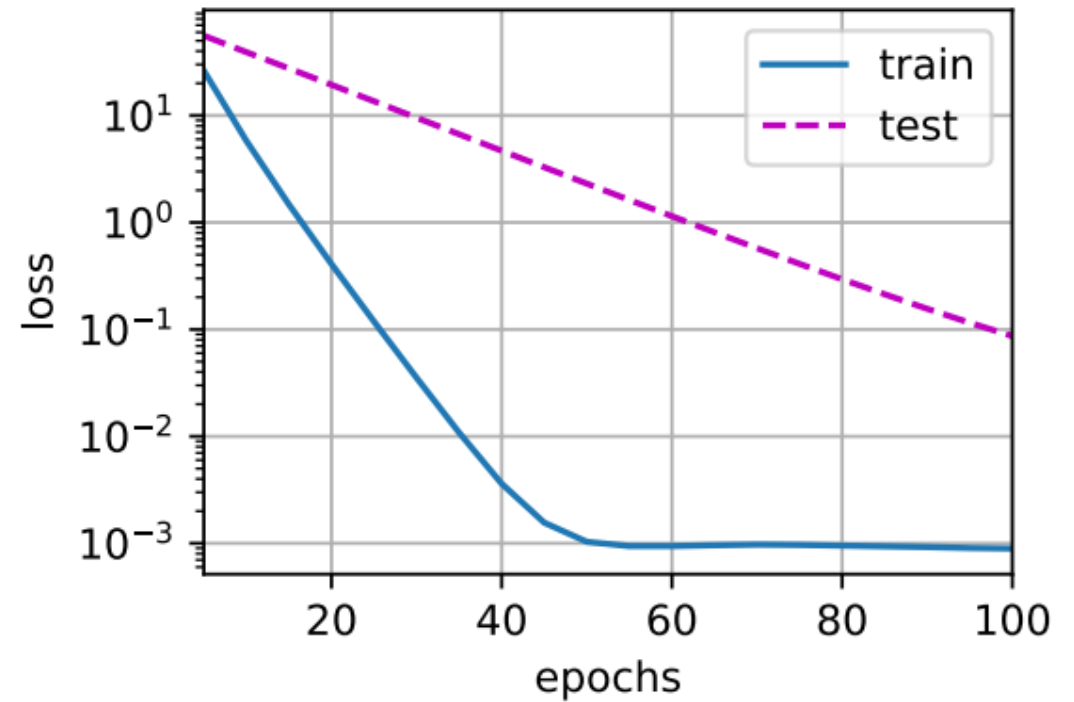
$$\text{Regularized Loss} = \text{Loss} + \lambda * \text{L2 Norm Penalty}$$

Bias-Variance Trade-Offs

Norm (W) = 12.9



Norm (W) = 0.3



Reference and Further Reading

- *FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS. Algorithms, Worked Examples, and Case Studies* John by D. Kelleher Brian Mac Namee Aoife D'Arcy (Permission taken MIT Press)
- *Dive into Deep Learning*, Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola (July 25, 2021)
- [Introduction to Multivariate Regression Analysis - PMC \(nih.gov\)](#)